

EXPERIMENT NO 1[B]

HALF -WAVE RECTIFIER WITH FILTER

Objective:

To examine the input and output waveforms of half wave Rectifier with filter and also Calculate its ripple factor.

Apparatus/Equipments:

S.No.	Apparatus	Specification
1	P-N Diode	IN4007
2	Resistor	1K Ω , 10K Ω
3	Digital Multimeter	
4	Transformer	(6V-0-6V)
5	Capacitor	100 μ f/4.7 μ f
6	DSO and DSO probes	
7	Bread board	
8	Connecting wires	
9	Function Generator	

THEORY:

In Half Wave Rectification, When AC supply is applied at the input, only Positive Half Cycle appears across the load whereas, the negative Half Cycle is suppressed. During positive half-cycle of the input voltage, the diode D is in forward bias and conducts through the load resistor R_L . Hence the current produces an output voltage across the load resistor R_L , which has the same shape as the +ve half cycle of the input voltage.

During the negative half-cycle of the input voltage, the diode is reverse biased and there is no current through the circuit. i.e., the voltage across R_L is zero. The net result is that only the +ve half cycle of the input voltage appears across the load. The average value of the half wave rectified o/p voltage is the value measured on dc voltmeter. The efficiency of the Half Wave Rectifier is 40.6 %.

Rectifiers with Filter: All rectifier outputs contain considerable amount of ripple in addition to the DC component. In order to avoid AC components, a filter is connected at the output of the rectifier.

The capacitor input filter is a simple and cost-effective filter. A high-value capacitor C is connected in parallel with the load resistor R_L . The capacitor charges to the peak voltage V_m during the positive half-cycle of the AC input.

After reaching the peak, the capacitor discharges slowly through R_L as the diode becomes reverse-biased. Before the capacitor voltage drops significantly, the next cycle arrives, and the capacitor recharges to the peak voltage.

The rms value of the filtered output is calculated assuming that the wave as a triangular wave and it is

$V_{r, rms} = V_{r, pp} / 2\sqrt{3}$, where $V_{r, pp}$ is the peak to peak value of the ripple voltage.
 $V_{dc} = V_m - (V_{r, pp}/2)$, Ripple factor $r = V_{r, rms}/V_{dc}$

For a half wave rectifier the ripple factor is also expressed as a function of capacitance and load resistance, $r = 1/2\sqrt{3}fR_L C$.

Q. Why is the capacitor connected in parallel and the inductor in series with the load?

CIRCUIT DIAGRAM:

Half Wave Rectifier With Filter:

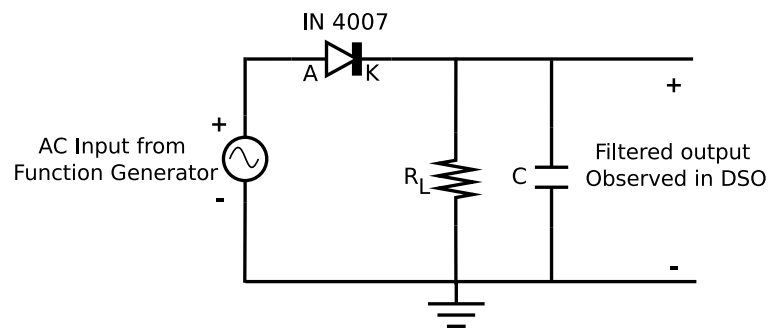


Figure 1: Half Wave Rectifier With Filter

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.
3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output of the filter.
4. Find the theoretical value of dc voltage by using the formula, $V_{dc} = V_m/\pi$
 Where, $V_m = 2V_{rms}$, (V_{rms} =output ac voltage.)
5. The Ripple factor is calculated by using the formula $r = \text{ac output voltage}/\text{dc output voltage}$

OBSERVATIONS AND CALCULATIONS:

Half wave rectifier (with filter)

V _{pp} (input voltage) (V)	V _{r,pp} (peak to peak ripple voltage) (V)	V _m (Peak volt- age) = V _{pp} /2 (V)	V _{dc} = V _m - V _{r,pp} /2 (V)	V _{rms} = V _{r,pp} /2√3 (V)	Ripple factor	
					Theoretical	Experimental

Table 1: Voltage and Ripple Factor Measurements

Theoretical calculation (with filter) :

$$\text{Ripplefactor} = 1/(2\sqrt{3}fR_LC) =$$

Experimental calculation (with filter) :

$$\text{Ripplefactor} = V_{rms}/V_{dc} =$$

MODEL WAVEFORMS

Half-Wave Rectifier with capacitor filter waveform

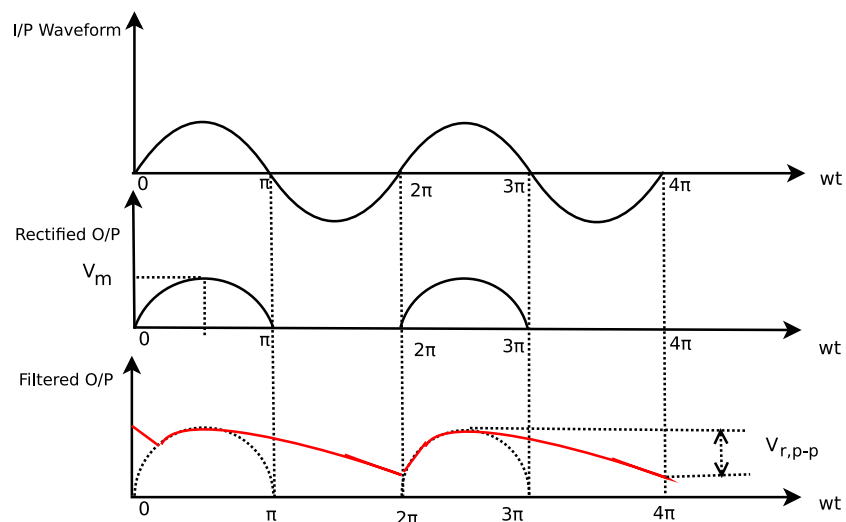


Figure 2: Half-Wave Rectifier with capacitor filter waveform

PRECAUTIONS:

1. The primary and secondary side of the transformer should be carefully identified.
2. The polarities of all the diodes should be carefully identified.

Results:

We have observed the Input and Output waveform of half wave rectifier with and without filter in DSO and also calculated experimental value of ripple factor of half wave rectifier with filter which is _____